

Robot-assisted Recognition of Vocal Emotions in Pseudospeech for Cochlear Implanted Adolescents

Gloria Araiza-Illan¹, Luke Meyer¹, Bert Maat¹, Deniz Başkent¹

¹Department of Otorhinolaryngology / Head and Neck Surgery, University Medical Centre Groningen, University of Groningen, The Netherlands

g.a.araiza.illan@rug.nl, l.meyer@rug.nl, a.maat@umcg.nl, d.baskent@rug.nl

Abstract

Recognising vocal emotions in speech is difficult for children with hearing loss and who use a cochlear implant (CI). As regular monitoring could be burdensome, we propose a NAO robot as a test interface. Adolescents with CIs (10-17yr) performed the EmoHI test for recognising vocal emotions in pseudospeech (no linguistic emotion information), once with a computer and once with a NAO. Interfaces are compared via test results and durations, and participants' perception of the interfaces. Test results (sensitivity index, d') were similar (0.36 ± 0.36 vs. 0.37 ± 0.43), but durations were significantly longer on the NAO (4.18 min $\pm$ 39 sec vs. 5.13 min $\pm$ 51 sec). The computer had a higher perceived usability, but the NAO was rated more enjoyable, engaging and preferable. Overall, the NAO sound quality seems sufficient for conducting the EmoHI test, even with a CI. The higher ratings of enjoyability for NAO may be especially useful in conducting such tests in populations with hearing devices.

Index Terms: vocal emotion recognition, cochlear-implanted adolescents, human-robot interaction, psychophysics, user-experience.

1. Introduction

Recognition of emotions in speech is vital for correctly understanding and interpreting the meaning, which is essential for social communication. Vocal emotion recognition relies on the perception and processing of relevant prosodic cues [1]. In hearing loss, such cues may not be sufficiently audible, or even altered due to suprathreshold distortions [2], or in cochlear implants, due to limitations of electric hearing [3], [4]. As the correct interpretation of the emotion prosodic cues would rely on use of past learning experiences, vocal emotion recognition in children, even those with normal-hearing, can take many years to develop to reach adult-like performance [5], [6], [7]. Hearing aids (HAs) and cochlear implants (CIs) provide compensation for hearing loss; however, it is not clear to what degree such compensation would be possible for vocal emotion recognition, or if such compensation would be possible for all device users [5], [6], [7], [8], [9]. For example, while children with CIs seem to have higher scores in voice or speech perception tests than adult CI users [10], they seem to have difficulty in vocal emotion recognition [5], [6].

Identifying potential difficulties and delays in vocal emotion recognition in children with hearing devices may require regular monitoring, which can be burdensome. To help maintain attention, concentration, and engagement, some computer interfaces use cartoons and animations, as well as a game-like progression and rewards [6], [11]. Socially assistive robots can also provide an effective interface to enhance participant's engagement and even performance during a task,

whilst offering a positive experience [12], [13], [14]. Literature has shown that the physical embodiment of an agent is preferred over its virtual counterpart, and can contribute favourably to its social presence [15], [16], potentially motivating interactors to exert more effort on a given task [17]. In areas such as vocal emotion recognition, which seems challenging for children and adults with hearing loss, such factors could be crucial.

Humanoid robots, such as the NAO from Aldebaran Robotics, are an example of socially assistive robots that are often used in areas such as education [18], and emotional and developmental therapy [19], [20]. The NAO has been shown to facilitate various learning applications in human-robot interactions due its small size, and friendly and human-like appearance [21]. The aim of this study is to evaluate the feasibility of the NAO robot to conduct a vocal emotion recognition test with adolescents with CIs.

2. Methods and materials

2.1. Participants

Forty-one Spanish speaking adolescents (10 - 17 yr; 15.16 ± 1.96 yr; 8 identified as female, 11 as male, and 22 preferred not to say) took the EmoHI test twice, one after the other (computer and robot, Figure 1). All participants were attendees of a Spanish summer camp for CI adolescents. Data from five participants were excluded for the analysis of the audiological component due to either missing or incomplete audio recording of their interaction with the robot, during which the vocal emotion recognition test was conducted. The audio recordings were used for the corroborating of spoken responses from participants. The legal guardians of the adolescents provided their written informed consent prior to the start of the study.

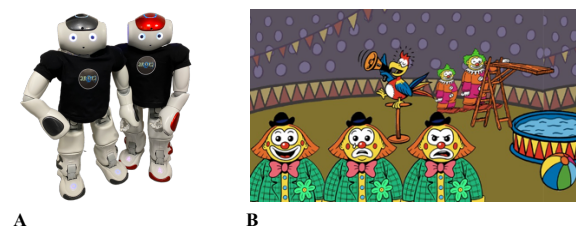


Figure 1: *EmoHI* test interfaces. NAO robots (panel A), child-friendly computer interface (panel B).

2.2. Stimuli

The EmoHI corpus consists of a closed set of pseudospeech stimuli with three emotional intonations (happy, angry, and sad), produced by four (two female and two male) native Dutch speakers [6]. The two pseudospeech sentences used are “Koun se mina lod belam” [kʌun sɔmina: lod be:lɑm] and “Nekal ibam

soud molen” [ne:kal ibam saut mo:lən] (from the Geneva Multimodal Emotion Portrayal (GEMEP) Corpus; [22]). While these phrases are plausible sound combinations for Western languages, they have no meaning in Indo-European languages, hence the participants cannot make use of the linguistic content for identifying the speech emotions. The number of presented stimuli totalled to 36; four speakers x three emotions x three sentence productions (one sentence repeated once + the other sentence).

Stimulus presentation level for all interfaces was calibrated to 75 dB SPL, using a Knowles Electronics Mannequin for Acoustic Research (KEMAR, GRAS, Holte, Denmark) head assembly and a Svantek sound level meter (Svan 979).

2.3. NAO setup

Two humanoid NAO robots were used in parallel, a V5 H25 and a V6. We named them “Sam” and “Robin”, gender-neutral names to elude a priori gender assignment of the robots.

The original EmoHI MATLAB code was rewritten into Python, allowing the test and stimuli to be run and stored directly on the robots. The stimuli were played using the onboard soundcard and through the internal stereo loudspeakers on either side of the robots’ heads. One of the preprogrammed voices available with the robots was used for when the robots talked, which differed from the voices of the test stimuli. This difference was explained to participants.

The experimental design was based on the “Wizard of Oz” (WoZ) technique [23], in which the robot is controlled by a human supervisor, unbeknownst to the participant interacting with the robot. This allowed participants to communicate with robots using speech, as previous studies reported people prefer to interact with robots via verbal communication [24], [25].

The interaction (all in Spanish) started with an introductory component, which included a short conversation with the participant before the main test explanation. The conversation consisted of a series of simple questions, including “what is your name?”, and “how are you doing today?”. Following the conversation, the test was explained and commenced. At the end of the test, another simple conversation was added, inquiring whether or not participants enjoyed the test and thanking them for their participation. Participants were asked to give their responses verbally to each of the presented questions and stimuli (naming the emotion). The human supervisor (“wizard”) manually logged the spoken responses of participants, both during the conversation and the testing components of the interaction. In order to allow for a “wizard” to log participants’ verbal responses, and to have control over multiple robot actions in parallel (e.g., moving the hands and body, blinking, speaking, presenting stimuli and giving feedback), the test implementation was redesigned to use the Robot Operating System (ROS). The full duration of the interaction was audio recorded for the offline (post-study) analysis of possible human error due to the WoZ implementation.

2.4. Computer setup

An HP Pavilion X360 and a Microsoft Surface Pro with Windows 10 and MATLAB 2022b were used in parallel. Stimuli were played through the computers’ internal sound cards, and a set of Logitech Z200 (Lausanne, Switzerland) external loudspeakers. Participants logged their responses using an external mouse connected to each computer.

The MATLAB computer interface (Figure 1, panel B) corresponds to a child-friendly circus-themed animation (as a serious game) composed of three cartoon clowns, each of which shows a specific facial expression: happy, angry, and sad. A parrot in the centre of the screen presented the stimuli, after which the participant clicked on the clown with the facial expression that matched the heard emotion of the voice.

2.5. Evaluation of the interface perception

Two questionnaires were used to assess how participants perceived the usability of both the NAO robot and the computer, and which interface (either robot or computer) the participants preferred based on their user experience (UX).

2.5.1. Usability questionnaire

The System Usability Scale (SUS) [26] is widely used to assess the perceived appropriateness of the proposed purpose of any interface [27]. The original SUS consists of 10 items, alternating between positive (odd items) and negative (even items), rated using a 5-point Likert scale. We used a shortened version of the Spanish translated SUS (hereafter referred to as child-SUS), based on the recommendation of two native Spanish speaking speech and language therapists, as three items were found to be too complex for child participants. Table 1 shows the English version of the child-SUS.

Table 1: *Child-SUS. In each item, [interface] is replaced with either “robot” or “computer”.*

7-item child-SUS
I think I would like to use the [interface] frequently.
I thought the [interface] was easy to use.
I think that I would need the support of a technical person to be able to use the [interface].
I would imagine that most people would learn to use the [interface] quickly.
I found the [interface] very cumbersome to use.
I felt confident using the [interface].
I needed to learn a lot of things before I could get going with the [interface].

In SUS, each item is scored between 0 and 4. For the odd numbered items, this corresponds to the Likert rating minus 1. For the even numbered items, which are reverse coded, this corresponds to 5 minus the Likert rating [26]. In the original SUS, these item scores are summed and multiplied by 2.5 to obtain a final SUS score out of 100. For the child-SUS, the scores are summed and multiplied by 3.57 to obtain the final SUS score out of 100. Furthermore, we also computed the corresponding SUS curved grading scale (CGS). The CGS is also consistent with industrial practices, and has become common that a score of at least 80 (grade of A-) is indicative of above average positive user experience [27].

2.5.2. User experience questionnaire

ISO 9241-210 [28] defines UX as “a person’s perceptions and responses resulting from the use or anticipated use of a product, system, or service”. We assessed participants’ UX via a 5-item questionnaire that required a binary response (selecting either the computer or the robot). The responses to each of the five items in the questionnaire are reported individually.

2.6. Experimental setup and procedure

The study was conducted at a summer camp house in Spain. Interfaces were placed at four different rooms and participants sat at a table facing each interface. The order of the test interface was randomised, counter-balancing potential learning and fatigue effects. Participants were first trained with four practice stimuli, randomly taken from the 36 stimuli. During the test, participants were presented with all 36 stimuli in random order and were instructed to indicate the recognised speech vocal emotion (happy, angry, or sad).

Positive and negative visual feedback were presented during training and data collection. For the computer, confetti was displayed for correct responses and a headshake from the parrot for incorrect responses. Similarly, the robot nodded when a correct response was given, and shook its head when an incorrect response was given.

After each EmoHI test, participants filled in the child-SUS for that specific interface. At the end of the entire experimental session, participants filled in the UX questionnaire, evaluating their preferences between the two interfaces.

2.7. Data analysis

Participants' sensitivity index d' (prime) was calculated per emotion per interface based on hit rates and false alarms [29]. In this study, the maximum possible d' is 2.77 for the 3-alternative-forced choice model, which corresponds to 91.67% maximum hit-rate; and 8.33% maximum false-alarms [29]. In interpreting d' , increasingly positive scores indicate a stronger probability of correct recognitions (hits), scores near zero indicate random recognitions and decreasing negative scores indicate a high probability of incorrect recognitions (false alarms), i.e., a systematic incorrect categorization. d' scores were used in a 2-by-2 repeated-measures ANOVA with two within-subject factors (two interfaces $\times$ three emotions). Where sphericity was violated, Greenhouse-Geisser corrections were applied. Bayesian repeated-measures ANOVA was also performed using the same factors described above, motivation for which is further discussed in [30]. In addition, d' scores were averaged across all emotions per interface and hence referred to as "overall/combined vocal emotion d' ". As this removed the factor of emotion, a paired-sample t-test (classical and Bayesian) was performed between these averages for either interface. To examine the effect of the interface on the test duration, classical and Bayesian t-tests were performed. Finally, paired-sample t-tests were also performed for each of the SUS items based on their raw Likert values between the computer and NAO interfaces.

3. Results

3.1. Vocal emotion recognition test results

Table 2 and Figure 2 show participants' d' scores. Figure 2, panel A shows the results of the d' scores for the CI adolescents for each emotion for both interfaces, and panel B the overall/combined vocal emotion d' scores (averaged across all emotions) per interface. These violin plots indicate density curves, with more dense regions indicating the quartile frequencies.

Repeated-measures ANOVA results showed no statistically significant effect on the d' scores for the interface factor [$F(1,35) = 0.015$, $p = 0.90$, $n_2p = 4.26e-4$], but a significant effect for the emotion factor [$F(2,70) = 7.47$, $p = 0.001$, $n_2p =$

0.176], as well as a significant interaction between the interface and emotion [$F(2,70) = 7.11$, $p = 0.002$, $n_2p = 0.169$]. Post-hoc analyses using a Bonferroni correction showed that for the emotion factor, the observed significant effect was due to significant differences between "happy" and "sad" ($p_{\text{bonf}} < 0.013$), and "angry" and "sad" ($p_{\text{bonf}} < 0.002$). A paired-sample t-test showed no significant difference for overall/combined vocal emotion d' scores between the two interfaces [$t(35) = -0.122$, $p = 0.904$, Cohen's $d = -0.023$].

Table 2: Mean d' scores and standard deviations for each emotion and overall/combined vocal emotion per interface.

Category	d' mean $\pm$ SD	
	Computer	NAO robot
Happy	0.35 $\pm$ 0.39	0.28 $\pm$ 0.43
Angry	0.19 $\pm$ 0.37	0.38 $\pm$ 0.42
Sad	0.58 $\pm$ 0.59	0.46 $\pm$ 0.69
Overall/combined	0.36 $\pm$ 0.36	0.37 $\pm$ 0.43

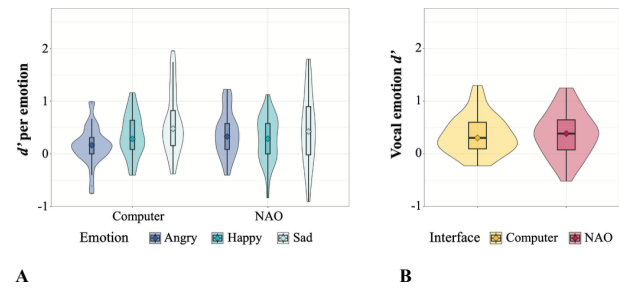


Figure 2: Box and violin plots of the d' scores per emotion (panel A) per interface and overall/combined vocal emotion d' score (panel B). Boxplots indicate minimum, maximum, median and interquartile ranges.

Bayesian repeated-measures ANOVA showed moderate evidence for no effect of the interface on the d' scores ($BF_{10} = 0.243$), strong evidence for an effect of the emotion ($BF_{10} = 25.529$), and strong evidence for an effect of the interaction between interface and emotion ($BF_{10} = 25.960$). Bayesian paired-sample t-test showed moderate evidence for no difference between the overall/combined vocal emotion d' scores between the interfaces ($BF_{10} = 0.180$).

3.2. Test duration

The test duration was 4.18 min $\pm$ 39 sec with the computer, and 5.13 min $\pm$ 51 sec with the NAO. Comparison of these durations showed the NAO was significantly longer than the computer [$t(34) = 6.12$, $p < 0.001$, Cohen's $d = 1.26$, $BF_{10} = 2.77e+4$].

3.3. Evaluation of the interface perception

Results of the child-SUS showed the computer was given a higher mean score (76.02 $\pm$ 16.04), and thus a higher CGS grade (B) than the NAO robot (63.01 $\pm$ 13.97, C-).

Results of the UX questionnaire (Figure 3) showed a preference for the NAO robot across all UX items, except for the item "Which was easier to hear?". The most preference was shown for the item "Which is more engaging?".

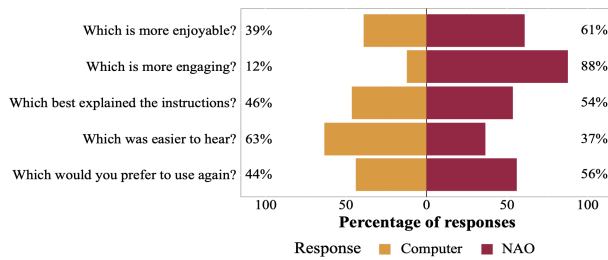


Figure 3: UX questionnaire results. Percentages show the proportion of participants who preferred the computer (left) or robot (right) interface per question.

4. Discussion

The outcome of the study showed both through classical and Bayesian inference methods that when a significant effect was present, it was due to the emotion factor, not the interface factor.

Regarding the significant effect of the emotion factor, as happiness and anger are both high arousal emotions, and share a number of vocal acoustic features, they are often confused. This was previously shown with adult CI users [31], and our results now confirm the patterns to be similar also in adolescent CI users. In contrast, sadness is low arousal, with distinct acoustic features than high arousal emotions, making it easier to recognise it in the EmoHI test, resulting in a higher d' score.

Regarding the interface factor, the present study expands the findings of previous literature comparing these two interfaces, but with other types of auditory and speech recognition tests, in typical-hearing adults, and in sound-treated rooms [13], [30]. The present study demonstrates the feasibility of the NAO robot as a test interface to be broader. The NAO can also be a feasible interface for assessing vocal emotion recognition, even when used in an uncontrolled and noisy environment (such as the summer camp), and even when tested with adolescents with CIs, despite the challenges of the developmental effects and hearing loss [5], [6].

The duration to complete the EmoHI test was significantly longer when using the NAO in comparison to the computer. An explanation for this may lie in the UX questionnaire results, where the majority of participants reported that the computer was easier to hear than the NAO. Due to the noisy conditions in which the NAOs were located at the summer camp, and the participants finding it difficult to hear the robots, it is possible that they might have had to focus more on throughout the test, taking longer to consider each of their responses.

Results of the child-SUS showed that overall perceived usability of the computer was greater than that for the NAO. NAO received a CGS grade four grades below than the computer. However, both interfaces received a grade lower than A-, the minimum SUS score for above average positive user experience [27]. This suggests that none of the interfaces are perceived by CI adolescents as highly appropriate for the proposed purpose (vocal emotion recognition). Furthermore, this might be linked to the difficulty that CI adolescents experience in recognising vocal emotions in general, as was explained in Introduction. The EmoHI test was designed to make the recognition task as doable as possible, by only using three core emotions, using stimuli that are short enough for test duration and long enough for expression of emotions, and by keeping the total test duration relatively short [6]. Despite this, perhaps there is still opportunity to improve and optimise the design of the EmoHI test to provide an increased positive

experience for participants, which could be realised with input from the adolescent CI users.

The higher child-SUS scores may depict a greater familiarity with a computer interface. However, this may in fact have the inverse effect regarding interface preference, supported by the majority of the adolescents preferring the robot over the computer in the UX questionnaire. Both the low child-SUS score and the general preference for the NAO may be linked to a novelty effect, given that the participants reported to not have interacted with the robot before. It is therefore possible that, through repeated interactions, as participants become more familiar with the NAO, its perceived usability may increase, but there is also the possibility that the preference for the NAO robot may decrease.

It is also possible that the child-SUS results were affected by the aforementioned non-ideal setups for the NAO robots. The locations of the robots (randomly chosen) were more open to the general ongoing of the camp, such as people entering the spaces or chatting relatively loudly nearby. Therefore, it may be that the computers appeared more usable, as participants were in much quieter conditions when using these interfaces.

Post-hoc characterisation of the speakers of all interfaces showed no evidence to expect that a difference in loudness and timbre of the stimuli would be perceived by participants.

Finally, regarding the implementation of the WoZ, it was hypothesised that for participants, verbally responding to the NAO would be a more natural and simple method of response logging, compared to using the robot's sensors. Not only does the verbal method greatly reduce the steps from recognition to response, but also minimises informational overload by limiting the processing system to only auditory information, excluding extra visual information, according to the multimedia learning theory [32]. Nevertheless, the WoZ potentially brought some limitations. This setup placed a great deal of reliance on the human supervisor ("wizard") to accurately record the responses of the participants (source of human error).

5. Conclusions

Our study showed the potential of a widely used socially assistive robot as an effective interactive vocal emotion recognition testing interface for CI adolescents, comparable to the standard computer interface (as a serious game). Taking into account that children and adolescents with hearing devices may need regular monitoring and testing, it is of clinical value to have varied, trustworthy and portable technological tools that could also offer a positive and engaging experience.

Further improvements to the proposed robot implementation e.g., using automatic speech recognition instead of a WoZ setup, may also increase the portability and accessibility of the proposed interface.

6. Acknowledgements

The study was approved by the University Medical Centre Groningen METc ethical review committee (METc 2018/427, ABR nr NL66549.042.18).

We thank Paolo Toffanin, Iris van Bommel, Evelien Birza, Jacqueline Libert, and Jop Luberti for their contribution to PICKA test battery; Federación de Asociaciones de Implantados Cocleares de España (AICE) for the invitation to conduct this study, and Javier Hurtado Oliva and María Elena León Bustos for collecting data.

7. References

- [1] R. Banse and K. R. Scherer, 'Acoustic profiles in vocal emotion expression', *J. Pers. Soc. Psychol.*, vol. 70, no. 3, pp. 614–636, 1996, doi: 10.1037/0022-3514.70.3.614.
- [2] B. C. J. Moore, 'Perceptual Consequences of Cochlear Hearing Loss and their Implications for the Design of Hearing Aids', *Ear Hear.*, vol. 17, no. 2, p. 133, Apr. 1996.
- [3] M. K. Everhardt, A. Sarampalis, M. Coler, D. Başkent, and W. Lowie, 'Meta-Analysis on the Identification of Linguistic and Emotional Prosody in Cochlear Implant Users and Vocoder Simulations', *Ear Hear.*, vol. 41, no. 5, p. 1092, Oct. 2020, doi: 10.1097/AUD.0000000000000863.
- [4] D. Başkent, E. Gaudrain, T. Tamati, and A. Wagner, 'Perception and psychoacoustics of speech in cochlear implant users', in *Scientific Foundations of Audiology: Perspectives from Physics, Biology, Modeling, and Medicine*, San Diego, CA: Plural Publishing, 2016.
- [5] M. Chatterjee et al., 'Voice emotion recognition by cochlear-implanted children and their normally-hearing peers', *Hear. Res.*, vol. 322, pp. 151–162, Apr. 2015, doi: 10.1016/j.heares.2014.10.003.
- [6] L. Nagels, E. Gaudrain, D. Vickers, M. Matos Lopes, P. Hendriks, and D. Başkent, 'Development of vocal emotion recognition in school-age children: the EmoHI test for hearing-impaired populations', *PeerJ*, vol. 8, p. e8773, 2020, doi: 10.7717/peerj.8773.
- [7] L. Rachman et al., 'Vocal-emotion recognition in school-age children with hearing aids', *Ear Hear.*, In press.
- [8] H. Goy, M. K. Pichora-Fuller, G. Singh, and F. A. Russo, 'Hearing Aids Benefit Recognition of Words in Emotional Speech but Not Emotion Identification', *Trends Hear.*, vol. 22, p. 2331216518801736, Jan. 2018, doi: 10.1177/2331216518801736.
- [9] X. Luo, Q.-J. Fu, and J. J. Galvin, 'Cochlear Implants Special Issue Article: Vocal Emotion Recognition by Normal-Hearing Listeners and Cochlear Implant Users', *Trends Amplif.*, vol. 11, no. 4, pp. 301–315, Dec. 2007, doi: 10.1177/1084713807305301.
- [10] L. Nagels, E. Gaudrain, D. Vickers, P. Hendriks, and D. Başkent, 'Prelingually Deaf Children With Cochlear Implants Show Better Perception of Voice Cues and Speech in Competing Speech Than Postlingually Deaf Adults With Cochlear Implants', *Ear Hear.*, vol. 45, no. 4, pp. 952–968, Apr. 2024, doi: 10.1097/AUD.0000000000001489.
- [11] E. E. Harding et al., 'Musical Emotion Categorization with Vocoder of Varying Temporal and Spectral Content', *Trends Hear.*, vol. 27, p. 23312165221141142, Jan. 2023, doi: 10.1177/23312165221141142.
- [12] M. J. Matarić and B. Scassellati, 'Socially assistive robotics', in *Springer Handbook of Robotics*, B. Siciliano and O. Khatib, Eds., Cham: Springer International Publishing, 2016, pp. 1973–1994. doi: 10.1007/978-3-319-32552-1_73.
- [13] L. Meyer, G. Araiza-Illan, L. Rachman, E. Gaudrain, and D. Başkent, 'Evaluating speech-in-speech perception via a humanoid robot', *Front. Neurosci.*, vol. 18, p. 1293120, 2024, doi: 10.3389/fnins.2024.1293120.
- [14] A. Robaczewski, J. Bouchard, K. Bouchard, and S. Gaboury, 'Socially Assistive Robots: The Specific Case of the NAO', *Int. J. Soc. Robot.*, vol. 13, no. 4, pp. 795–831, Jul. 2021, doi: 10.1007/s12369-020-00664-7.
- [15] W. A. Bainbridge, J. Hart, E. S. Kim, and B. Scassellati, 'The effect of presence on human-robot interaction', in *RO-MAN 2008 - The 17th IEEE International Symposium on Robot and Human Interactive Communication*, Aug. 2008, pp. 701–706. doi: 10.1109/ROMAN.2008.4600749.
- [16] C. D. Kidd and C. Breazeal, 'Effect of a robot on user perceptions', in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (IEEE Cat. No.04CH37566), Sep. 2004, pp. 3559–3564 vol.4. doi: 10.1109/IROS.2004.1389967.
- [17] C. F. Bond, 'Social facilitation: A self-presentational view', *J. Pers. Soc. Psychol.*, vol. 42, no. 6, pp. 1042–1050, 1982, doi: 10.1037/0022-3514.42.6.1042.
- [18] A. K. Pandey and R. Gelin, 'Humanoid Robots in Education: A~Short~Review', in *Humanoid Robotics: A Reference*, A. Goswami and P. Vadakkepat, Eds., Dordrecht: Springer Netherlands, 2017, pp. 1–16. doi: 10.1007/978-94-007-7194-9_113-1.
- [19] K. Kinney et al., 'Speech characteristics of children with ASD to a humanoid robot and peers', *Proc. Meet. Acoust.*, vol. 45, no. 1, p. 060001, Jan. 2022, doi: 10.1121/2.0001519.
- [20] F. Yakub, A. Z. Md. Khudzari, and Y. Mori, 'Recent trends for practical rehabilitation robotics, current challenges and the future', *Int. J. Rehabil. Res.*, vol. 37, no. 1, p. 9, Mar. 2014, doi: 10.1097/MRR.0000000000000035.
- [21] A. Amirova, N. Rakhymbayeva, E. Yadollahi, A. Sandygulova, and W. Johal, '10 Years of Human-NAO Interaction Research: A Scoping Review', *Front. Robot. AI*, vol. 8, Nov. 2021, doi: 10.3389/frobt.2021.744526.
- [22] T. Bänziger, M. Mortillaro, and K. R. Scherer, 'Introducing the Geneva Multimodal expression corpus for experimental research on emotion perception', *Emot. Wash. DC*, vol. 12, no. 5, pp. 1161–1179, Oct. 2012, doi: 10.1037/a0025827.
- [23] L. Riek, 'Wizard of Oz Studies in HRI: A Systematic Review and New Reporting Guidelines', *J. Hum.-Robot Interact.*, pp. 119–136, Aug. 2012, doi: 10.5898/JHRI.1.1.Riek.
- [24] K. Dautenhahn, S. Woods, C. Kaouri, M. L. Walters, K. L. Koay, and I. Werry, 'What is a robot companion - friend, assistant or butler?', in *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Aug. 2005, pp. 1192–1197. doi: 10.1109/IROS.2005.1545189.
- [25] C. Ray, F. Mondada, and R. Siegwart, 'What do people expect from robots?', in *2008 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Sep. 2008, pp. 3816–3821. doi: 10.1109/IROS.2008.4650714.
- [26] J. Brooke, 'SUS: A "Quick and Dirty" Usability Scale', in *Usability Evaluation In Industry*, CRC Press, 1996.
- [27] J. R. Lewis, 'The System Usability Scale: Past, Present, and Future', *Int. J. Human-Computer Interact.*, vol. 34, no. 7, pp. 577–590, Jul. 2018, doi: 10.1080/10447318.2018.1455307.
- [28] ISO Central Secretary, 'Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems'. Accessed: Apr. 09, 2024. [Online]. Available: <https://www.iso.org/obp/ui/#iso:std:iso:9241:-210:ed-2:vl:en>
- [29] C. D. Creelman and N. A. Macmillan, *Detection Theory: A User's Guide*, 2nd ed. New York: Psychology Press, 2004. doi: 10.4324/9781410611147.
- [30] L. Meyer, L. Rachman, G. Araiza-Illan, E. Gaudrain, and D. Başkent, 'Use of a humanoid robot for auditory psychophysical testing', *PLOS ONE*, vol. 18, no. 12, p. e0294328, Dec. 2023, doi: 10.1371/journal.pone.0294328.
- [31] E. Harding et al., 'EXPRESS: Vocal and musical emotion perception, voice cue discrimination, and quality of life in cochlear implant users with and without acoustic hearing', *Q. J. Exp. Psychol.*, p. 17470218251316499, Jan. 2025, doi: 10.1177/17470218251316499.
- [32] R. E. Mayer and R. Moreno, 'A cognitive theory of multimedia learning: Implications for design principles', *J. Educ. Psychol.*, vol. 91, no. 2, pp. 358–368, 1998.